

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA1003	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
Student Name:	G.Mohammed Shariff	Reg. No.:	192425378
Date:	22/07/2026	Duration:	60 minutes

Code of Conduct

I G.Mohammed Shariff/192425378 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **MOHAMMED SHARIFF**

Annexure A – Case Study Analysis

TITLE:AI-Based Learning Management System with Personalized Learning Recommendations

DOMAIN:EDUCATION

1. Understand the Problem Statement

1.1 Explain the problem in your own words

Traditional Learning Management Systems (LMS) primarily function as repositories for course materials, assignments, and assessments. Although they enable digital learning, they generally adopt a **one-size-fits-all** approach by providing identical learning resources to every student, regardless of individual learning styles, academic performance, prior knowledge, or pace of learning. This limits the effectiveness of the learning process because students possess different strengths, weaknesses, and educational needs.

The proposed **AI-Based Learning Management System (AI-LMS)** addresses this limitation by integrating Artificial Intelligence and Machine Learning techniques to analyse learner behaviour, academic performance, assessment results, content interaction, and engagement patterns. Based on this analysis, the system generates **personalized learning recommendations**, including relevant study materials, practice exercises, revision plans, and suitable courses for each learner.

1.2 Identify the objectives of the proposed system

The primary objective of the proposed AI-Based Learning Management System is to develop an intelligent, adaptive, and user-centric educational platform capable of delivering personalized learning experiences while improving the overall efficiency and effectiveness of teaching and learning processes. The system aims to leverage Artificial Intelligence and Machine Learning techniques to analyse students learning behaviour, academic performance, preferences, and engagement patterns in order to recommend appropriate learning resources, assessments, and study plans tailored to individual needs

The specific objectives are:

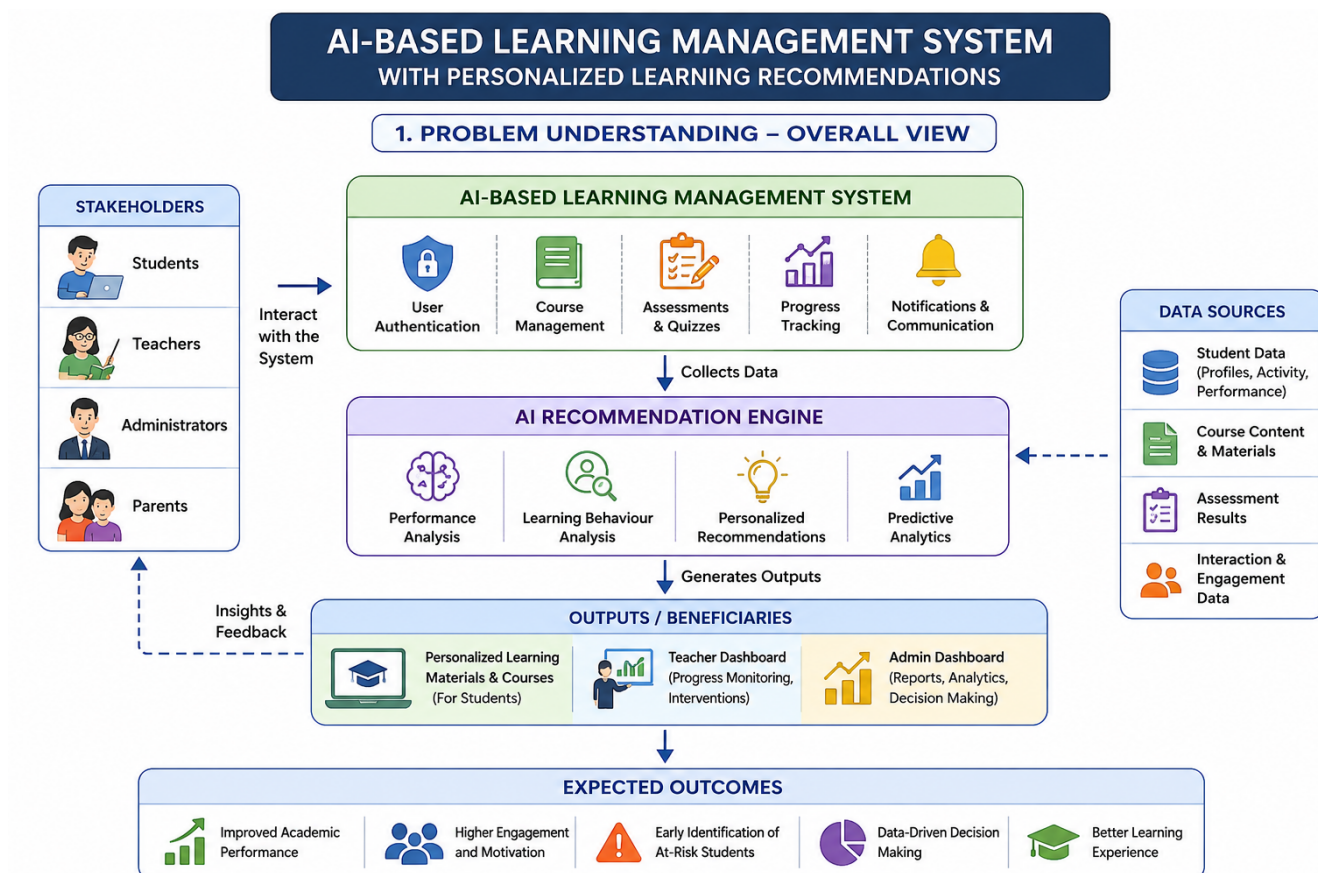
- Develop an AI-powered recommendation engine that suggests personalized learning resources based on individual student performance, learning behaviour, interests, and competency levels.
- Improve student engagement and academic achievement by delivering adaptive learning pathways instead of generic course content.
- Continuously monitor learner progress using performance analytics, assessment scores, attendance records, and interaction history.
- Assist educators in identifying weak-performing students through predictive analytics, enabling timely academic intervention.
- Automate routine academic processes such as content recommendation, progress tracking, assignment evaluation, notifications, and report generation.
- Provide administrators with comprehensive dashboards and analytical reports for effective academic planning, resource allocation, and institutional decision-making.
- Ensure secure management of student information while maintaining data privacy, integrity, and accessibility.
- Build a scalable and maintainable platform capable of supporting increasing numbers of users, courses, and educational resources.
- **To support data-driven decision-making** by providing administrators and educators with comprehensive dashboards, learning analytics, and performance reports for effective academic planning and institutional improvement.
- **To enhance collaboration and communication** among students, teachers, and parents through integrated messaging, notifications, feedback mechanisms, and progress updates, creating a more interactive learning environment.

1.3 State the scope of the problem

The scope of the proposed system covers the complete lifecycle of digital learning within educational institutions, including schools, colleges, universities, and online learning platforms.

The system provides the following functionalities:

- Student registration, authentication, and profile management.
- Course creation, content management, and curriculum organization.
- AI-driven personalized learning recommendations based on learner performance and behaviour.
- Online quizzes, assignments, examinations, and automated evaluation.
- Real-time performance monitoring through interactive dashboards and learning analytics.
- Progress tracking, attendance management, and academic reporting.
- Teacher dashboards for monitoring class performance and identifying learning difficulties.
- Administrative modules for user management, course administration, and system monitoring.
- Secure storage and management of educational data using role-based access control and encryption mechanisms.



2. Identify Key Stakeholders and Challenges

2.1 Identify all the stakeholders involved in the system

The successful implementation of an AI-Based Learning Management System involves multiple stakeholders who interact with the system throughout its lifecycle.

Primary Stakeholders

- Students
- Teachers/Faculty Members
- Educational Institution (School/College/University)

Secondary Stakeholders

- Parents/Guardians
- AI Recommendation Engine
- Database Administrator
- Software Developers

2.2 Explain the roles and responsibilities of each stakeholder

Students

Students are the primary users of the system. They register for courses, access learning materials, complete quizzes and assignments, receive AI-generated personalized recommendations, monitor their academic progress, and communicate with instructors.

Teachers/Faculty Members

Teachers manage courses, conduct assessments, monitor student performance, and provide personalized guidance using AI-driven insights.

Educational Institution

The institution defines academic policies, manages courses and departments, ensures the quality of education, allocates resources, and evaluates overall academic performance using institutional reports.

System Administrator

The administrator manages user accounts, system configuration, access control, database maintenance, server availability, software updates, security policies, backups, and disaster recovery.

Parents/Guardians

Parents can monitor their child's academic performance, attendance, learning progress, and receive notifications regarding important academic activities.

AI Recommendation Engine

The AI engine analyses student behaviour, learning patterns, assessment scores, and engagement data to generate intelligent recommendations that improve learning outcomes.

Software Development Team

Developers design, implement, maintain, and enhance the system while ensuring compliance with software engineering standards.

Quality Assurance Team

The QA team validates system functionality, performs software testing, identifies defects, and ensures that the software satisfies functional and non-functional requirements.

Cloud Service Provider

Provides scalable infrastructure, secure storage, high availability, backup services, and disaster recovery mechanisms.

2.3 Discuss the major challenges and issues faced by the stakeholders

Students

- Difficulty identifying appropriate learning resources.
- Different learning speeds and capabilities.

Teachers

- Monitoring hundreds of students manually.
- Identifying weak learners at an early stage.

Administrators

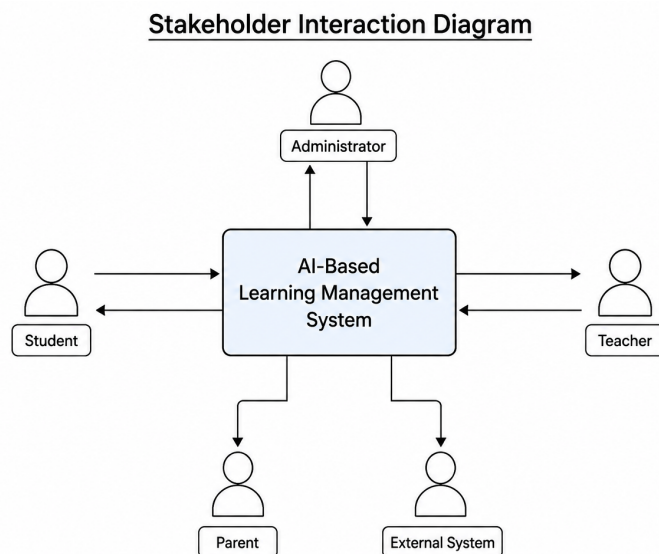
- Managing large volumes of educational data.
- Maintaining system security and privacy.
- Generating accurate institutional reports.

Parents

- Limited visibility into student performance.
- Delayed communication regarding academic progress.

Technical Challenges

- Protecting confidential educational data.
- Maintaining AI model accuracy.



3. Analyze the Current Approach

3.1 Describe the existing system or current process

Most educational institutions currently use conventional Learning Management Systems such as Moodle, Google Classroom, or Blackboard. These systems primarily serve as platforms for uploading study materials, conducting online assessments, managing assignments, and facilitating communication between students and instructors.

However, these systems largely follow a standardized approach where identical learning content is delivered to all students regardless of their academic strengths, weaknesses, learning pace, or interests. Teachers manually analyse student performance and identify learning difficulties, making the process time-consuming and less efficient.

3.2 Identify its strengths and limitations

Strengths

- Digital storage of educational resources.
- Easy access to learning materials from anywhere.
- Online assignment submission.
- Automated attendance and grading.
- Improved communication between students and teachers.
- Centralized academic management.

Limitations

- No personalized learning recommendations.
- Limited use of Artificial Intelligence.
- Manual monitoring of student progress.
- Inability to predict academic risks.

- Generic content delivery reduces student engagement.
- Lack of adaptive learning pathways.
- Limited analytical insights for decision-making.

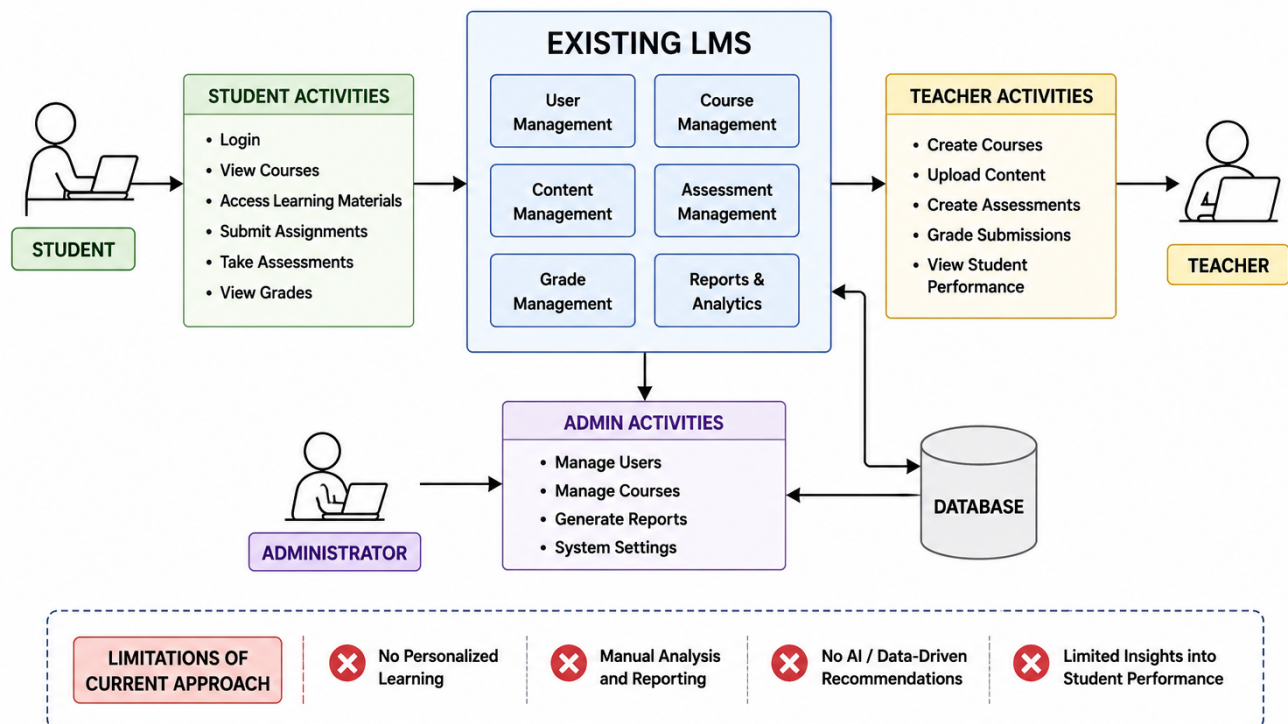
3.3 Analyze the problems or gaps that need to be addressed

The existing system has several shortcomings that reduce learning effectiveness.

- Lack of adaptive learning based on individual student performance.
- Inability to identify at-risk students early.
- Manual analysis consumes significant faculty time.
- Limited predictive analytics.
- Poor student engagement due to standardized learning.
- Absence of intelligent recommendation systems.
- Insufficient support for personalized academic planning.
- Limited scalability for advanced AI-based educational services.

3. ANALYZE THE CURRENT APPROACH

(Existing Traditional Learning Management System)



4. Propose a Suitable Solution Using Software Engineering Principles

4.1 Design a software-based solution for the given problem

The proposed solution is an **AI-Based Learning Management System (AI-LMS)** that combines cloud computing, machine learning, data analytics, and modern software engineering practices to create an intelligent learning environment.

The system continuously collects student interaction data, assessment scores, attendance records, learning preferences, and behavioural patterns. Using machine learning algorithms, it generates personalized learning recommendations, identifies weak concepts, predicts future academic performance, and provides adaptive study plans.

Teachers receive real-time dashboards that help them monitor individual and class performance, while administrators gain access to institutional analytics for informed decision-making.

4.2 Describe the major functional modules of the proposed system

User Authentication Module

Secure login, registration, password management, and role-based access control.

Student Management Module

Student profiles, enrollment, attendance, academic records, and learning history.

Course Management Module

Course creation, content upload, syllabus management, scheduling, and resource organization.

AI Recommendation Engine

Analyses learning behaviour and recommends personalized study materials, videos, quizzes, and revision plans.

Assessment Module

Online quizzes, assignments, examinations, automatic evaluation, and grading.

Learning Analytics Dashboard

Displays progress reports, performance trends, engagement metrics, and predictive insights.

Notification Module

Provides reminders for assignments, examinations, course updates, and personalized recommendations.

Administration Module

User management, role management, report generation, system monitoring, and institutional analytics.

4.3 Identify the functional and non-functional requirements

Functional Requirements

- User registration and authentication.
- Course enrollment and management.
- AI-based recommendation generation.
- Assessment creation and evaluation.
- Progress tracking.
- Report generation.
- Notification services.
- Dashboard visualization.
- Data backup and recovery.

Non-Functional Requirements

- High Availability
- Scalability
- Reliability
- Security
- Performance
- Maintainability
- Portability
- Interoperability

4.4 Explain how software engineering principles are incorporated into the solution

Modularity

The system uses independent modules for authentication, course management, AI recommendations, assessments, analytics, and administration, simplifying development, testing, and maintenance.

Scalability

Cloud-based architecture enables the system to efficiently support thousands of simultaneous users while maintaining consistent performance.

Maintainability

Clean architecture, reusable components, proper documentation, and modular coding practices simplify future updates and maintenance.

Reliability

Robust exception handling, automated backups, fault tolerance, and continuous monitoring ensure dependable system operation.

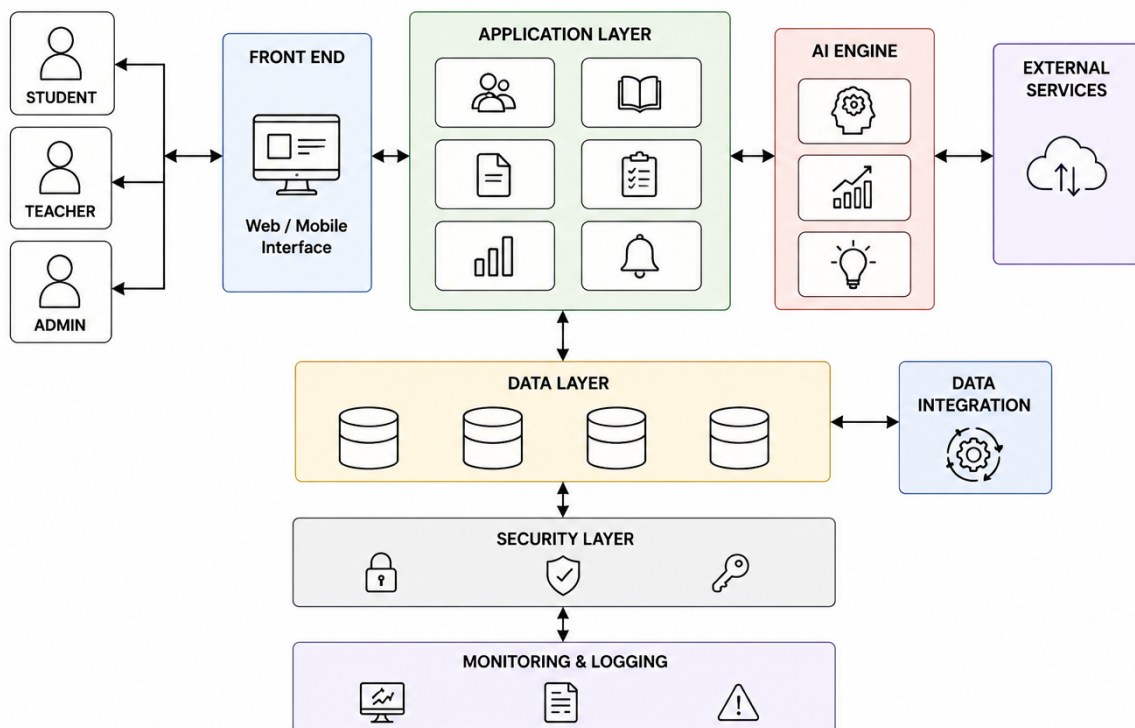
Usability

A responsive and intuitive interface improves accessibility for students, teachers, and administrators across desktop and mobile devices.

Security

The proposed system incorporates multiple security mechanisms to protect sensitive educational data and ensure secure system operation. It implements **role-based access control (RBAC)** to restrict access based on user roles, **encrypted password storage** using secure hashing algorithms, **HTTPS communication** to protect data transmitted over the network, and **secure APIs** for safe interaction between system components. Additionally, **database encryption**, **audit logging**, **regular security updates**, **automated backups**, and **multi-factor authentication (MFA)** help safeguard user information, prevent unauthorized access, ensure data integrity, and maintain compliance with institutional security and privacy standards.

4. PROPOSE A SUITABLE SOLUTION USING SOFTWARE ENGINEERING PRINCIPLES



5. Justify the Chosen Methodology/Framework

5.1 Select an appropriate SDLC model

Agile Software Development Methodology (Scrum Framework)

5.2 Justify why the selected methodology is suitable

The Agile Scrum methodology is the most appropriate choice because the AI-Based Learning Management System is a complex and evolving software product where user requirements are expected to change frequently.

The AI recommendation engine requires continuous refinement based on user feedback, educational outcomes, and evolving machine learning models. Agile supports iterative development, enabling frequent releases, continuous testing, stakeholder collaboration, and rapid incorporation of new requirements.

Unlike the Waterfall model, Agile allows flexible planning and incremental feature delivery, making it highly suitable for educational platforms where continuous improvement is essential.

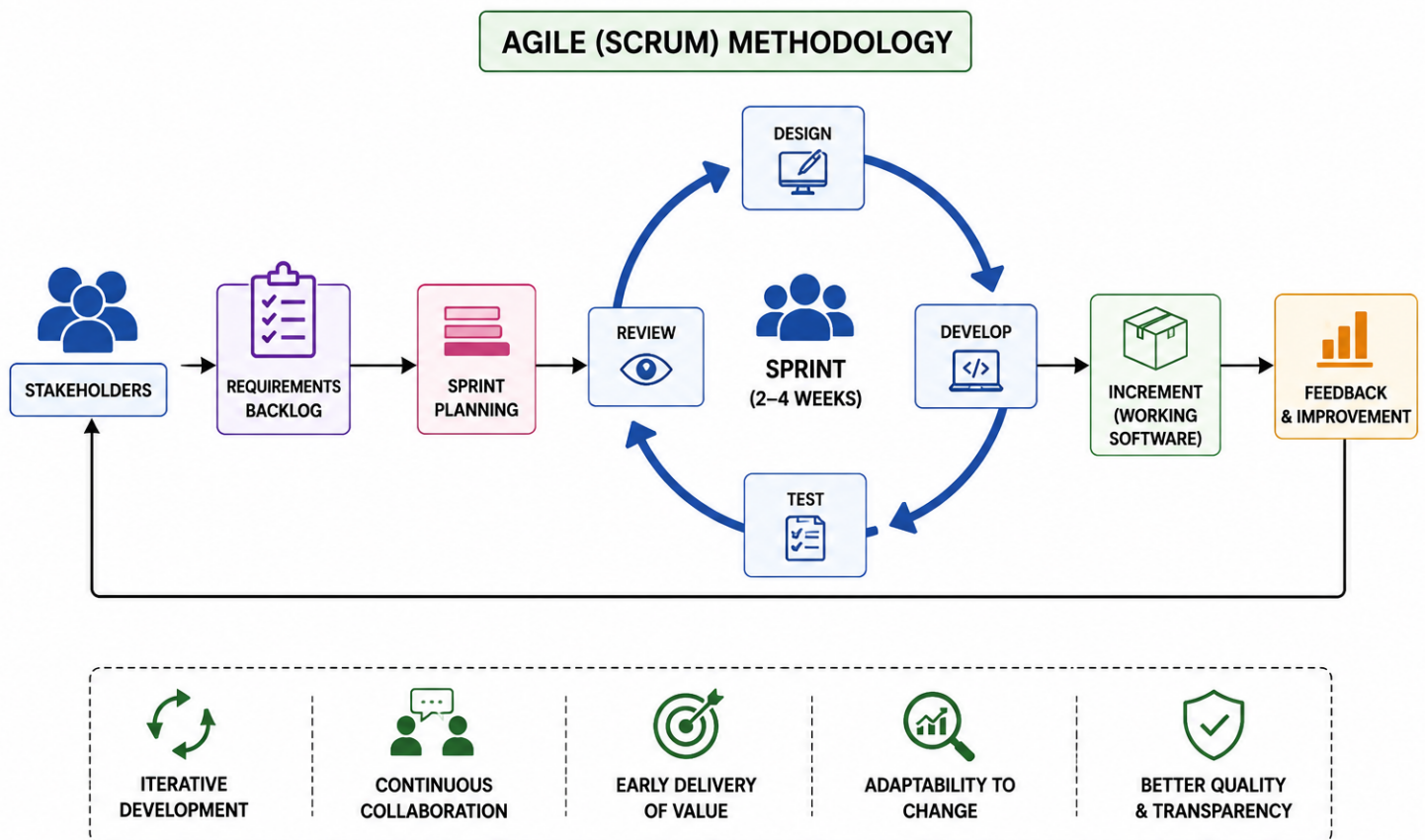
Additionally, Scrum ceremonies such as Sprint Planning, Daily Scrum, Sprint Review, and Sprint Retrospective facilitate effective communication among developers, stakeholders, educators, and institutional administrators.

5.3 Explain how Agile supports successful project development and maintenance

Agile improves software quality by promoting continuous integration, automated testing, iterative development, and frequent stakeholder feedback.

- Faster delivery of working software.
- Early identification and correction of defects.
- Continuous enhancement of AI algorithms.
- Better risk management.
- Improved software quality.
- Reduced development cost.
- Easier maintenance through modular releases.
- Continuous adaptation to changing educational requirements.

5. JUSTIFY THE CHOSEN METHODOLOGY / FRAMEWORK



6. Discuss Expected Outcomes and Limitations

6.1 Describe the expected benefits of the proposed solution

The proposed AI-Based Learning Management System is expected to significantly improve educational quality by delivering personalized learning experiences. Students will benefit from adaptive learning paths tailored to their individual strengths and weaknesses, leading to higher engagement, improved knowledge retention, and better academic performance. Teachers will gain real-time insights into student progress, enabling timely interventions and reducing manual monitoring efforts. Educational institutions will benefit from comprehensive analytics, efficient resource management, and informed decision-making. Overall, the system enhances learning efficiency, automates routine academic processes, and supports a more data-driven educational environment.

6.2 Explain how the solution addresses the identified challenges

The proposed system addresses the shortcomings of traditional Learning Management Systems by integrating AI-driven personalization and predictive analytics. Instead of delivering identical content to all learners, the AI engine analyses each student's learning behaviour, assessment results, and interaction history to recommend appropriate study materials and adaptive learning paths. Real-time dashboards enable teachers to identify struggling students early, while automated progress tracking reduces administrative workload. Advanced analytics provide institutions with actionable insights, improving academic planning and overall educational effectiveness.

6.3 Discuss potential risks, implementation challenges, and limitations

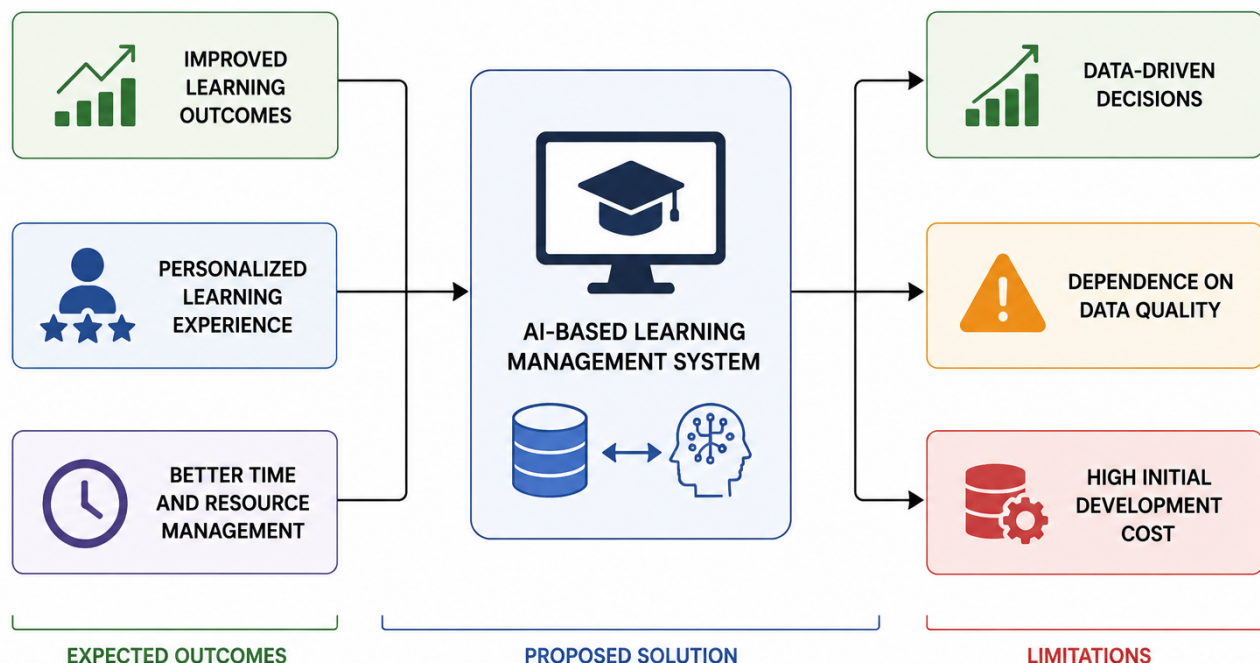
Despite its advantages, the system faces several challenges. Developing and maintaining AI models requires high-quality data, significant computational resources, and specialized technical expertise. Protecting sensitive student information is critical, making compliance with data privacy regulations and robust cybersecurity measures essential. The initial deployment may involve substantial infrastructure

and implementation costs, and resistance to adopting new technologies from users may slow implementation. Additionally, AI recommendations may be less effective if historical data is incomplete or biased, requiring continuous model training and validation.

6.4 Suggest possible future enhancements

Future enhancements can further improve the intelligence and effectiveness of the system by incorporating advanced technologies. These include AI-powered virtual tutors capable of providing real-time academic assistance, natural language processing for intelligent chatbots, predictive analytics to identify students at risk of poor performance, and adaptive assessments that automatically adjust question difficulty based on learner ability. Additional improvements may include multilingual support, mobile application development, gamification features to increase engagement, integration with virtual reality (VR) and augmented reality (AR) learning environments, blockchain-based academic credential verification, and learning analytics powered by deep learning algorithms.

6. DISCUSS EXPECTED OUTCOMES AND LIMITATIONS



Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.

Methodology/ Framework Justification	Selects the most suitable SDLC/ model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.
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Criteria	Mark
Problem Understanding	/5
Stakeholder & Challenge Analysis	/5
Current System Analysis	/5
Proposed Software Solution & Software Engineering Principles	/5
Methodology/ Framework Justification	/5
TOTAL	/5